2-1.

(1). $FT[x(n-n_0)] = X(e^{j\omega})e^{-jn_0\omega}$

(2) $FT[x^*(n)] = X^*(e^{-j\omega})$

(3) $FT[x(-n)] = X(e^{-j\omega})$

(4) $FT[x(n)*y(n)] = X(e^{j\omega})Y(e^{j\omega})$

(5) $FT[x(n)\cdot y(n)] = \frac{1}{2\pi}X(e^{j\omega})*Y(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\omega'})Y(e^{j(\omega-\omega')})d\omega'$

(6) $FT[n\,x(n)] = j\frac{dX(e^{j\omega})}{d\omega}$

(7)

(8) $FT[x^2(n)] = \frac{1}{2\pi}X(e^{j\omega})*X(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\omega'})X(e^{j(\omega-\omega')})d\omega'$

(9)

$x_9(n) = \sum_{n=-\infty}^{\infty}x(\frac{n}{2})$，其中n为偶数

令 $n' = \frac{n}{2}$

则 $FT(x_9(n)) = \sum_{n'=-\infty}^{\infty}x(n')e^{-j2\omega n'} = X(e^{j2\omega})$

(7)

(7) $x(2n) = \frac{1}{2}[(-1)^n+1]x(n)$

$ZT[x(2n)] = \frac{1}{2}[X(\sqrt{z})+X(-\sqrt{z})]$

令 $z = e^{j\omega}$

不 $FT(x(2n)) = \frac{1}{2}[X(e^{j\frac{\omega}{2}})+X(e^{-j\frac{\omega}{2}})]$

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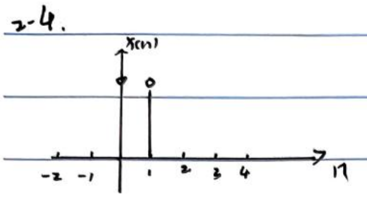
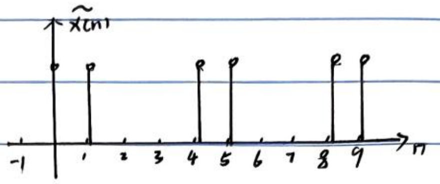
2-4.





$\tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}(n)\, e^{-j\frac{2\pi}{N}kn}$ ，其中 $N=4$

$= \sum_{n=0}^{1} e^{-j\frac{2\pi}{4}kn}$

$= 1 + e^{-j\frac{2\pi}{4}k} = 1 + e^{-j\frac{\pi}{2}k}$

$= e^{-j\frac{2\pi}{N}\frac{k}{2}} \frac{\sin(\frac{2\pi}{N}k)}{\sin(\frac{\pi}{N}k)}$

$= e^{-j\frac{\pi}{N}k} \frac{\sin(\frac{2\pi}{N}k)}{\sin(\frac{\pi}{N}k)} = e^{-j\frac{\pi}{4}k} \frac{\sin(\frac{\pi}{2}k)}{\sin(\frac{\pi}{4}k)}$ $\tilde{X}(k)$以4为周期.

$X(e^{jw}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \tilde{X}(k)\, \delta(w - \frac{2\pi}{N}k)$

$= \frac{\pi}{2} \sum_{k=-\infty}^{\infty} \tilde{X}(k)\, \delta(w - \frac{\pi}{2}k)$

$= \frac{\pi}{2} \sum_{k=-\infty}^{\infty} e^{-j\frac{\pi}{4}k} \frac{\sin(\frac{\pi}{2}k)}{\sin(\frac{\pi}{4}k)}\, \delta(w - \frac{\pi}{2}k)$

2-5.

(1) $FT[x(n)] = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$

令 $\omega = 0$

则有 $FT[x(n)] = \sum_{n=-\infty}^{\infty} x(n) = X(e^{j0}) = \sum_{n=-3}^{7} x(n) = 6$

(2) $x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$

令 $n = 0$

$x(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$

∴ 原式 $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 2\pi x(0) = 4\pi$

(3) 由(1)可知

$X(e^{j\pi}) = \sum_{n=-3}^{7} x(n) e^{-jn\pi}$

$= \sum_{n=-3}^{7} x(n) \cos n\pi$

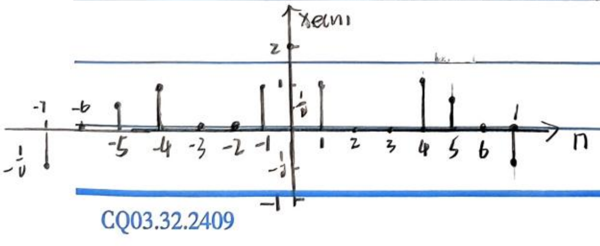
$= x(n)(-1)^n$

(4) $Re[X(e^{j\omega})] \Leftrightarrow x_a(n) = x_e(n)$

其中 $x_e(n) = \frac{1}{2}(x(n) + x^*(-n))$

又因 $x(n)$ 为实序列

则 $x_e(n) = \frac{1}{2}(x(n) + x(-n))$



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(5)

根据帕斯瓦尔

$$\sum_{n=-\infty}^{\infty}|x(n)|^2=\frac{1}{2\pi}\int_{-\pi}^{\pi}|X(e^{j\omega})|^2 d\omega$$

则 $\int_{-\pi}^{\pi}|X(e^{j\omega})|^2 d\omega=2\pi\sum_{n=-\infty}^{\infty}|x(n)|^2$

(6) $FT[nx(n)]=j\frac{dX(e^{j\omega})}{d\omega}$

则 $\int_{-\pi}^{\pi}\left|\frac{dX(e^{j\omega})}{d\omega}\right|^2 d\omega=2\pi\sum_{n=-\infty}^{\infty}|nx(n)|^2$

2-6

(1). $FT[x_1(n)]=e^{-j3\omega}$

(2) $FT[\delta(n)]=1$

$FT[x_2(n)]=\frac{1}{2}e^{j\omega}+\frac{1}{2}e^{-j\omega}+1$

$=\cos\omega+1$

(3) $FT[x_3(n)]=\frac{1}{1-ae^{-j\omega}}$

(4) $x_4(n)=u(n+3)-u(n-4)=R_7(n+3)$

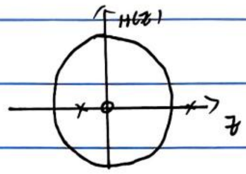
$FT(x_4(n))=e^{-j3\omega}\frac{\sin(\frac{7}{2}\omega)}{\sin(\frac{\omega}{2})}e^{+j3\omega}$

$=\frac{\sin(\frac{7}{2}\omega)}{\sin(\frac{\omega}{2})}$

2-23

(1) $y(n)-y(n-1)-y(n-2)=x(n-1)$

$$H(z)=\frac{z^{-1}}{1-z^{-1}-z^{-2}}=\frac{z}{z^2-z-1}\qquad z^2-z-1=0\ \Rightarrow\ \frac{1\pm\sqrt{5}}{2}$$



(2)

$$\frac{H(z)}{z}=\frac{1}{(z-z_1)(z-z_2)}=\frac{1}{\left[z-\left(\frac{1+\sqrt{5}}{2}\right)\right]\left[z-\left(\frac{1-\sqrt{5}}{2}\right)\right]}$$

$$=\frac{1}{\sqrt{5}}\ \frac{1}{z-\left(\frac{1+\sqrt{5}}{2}\right)}+\frac{-1}{z-\left(\frac{1-\sqrt{5}}{2}\right)}$$

$$H(z)=\frac{1}{\sqrt{5}}\left[\frac{z}{z-\left(\frac{1+\sqrt{5}}{2}\right)}-\frac{z}{z-\left(\frac{1-\sqrt{5}}{2}\right)}\right]$$

∵为因果 则 $|z|>\frac{1+\sqrt{5}}{2}$

$$h(n)=\frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n-\left(\frac{1-\sqrt{5}}{2}\right)^n\right]u(n)$$

(3) ∵ $H(z)$为稳定系统 则 $\left|\frac{1-\sqrt{5}}{2}\right|<|z|<\left|\frac{1+\sqrt{5}}{2}\right|$

$$h(n)=\frac{1}{\sqrt{5}}\left[-\left(\frac{1+\sqrt{5}}{2}\right)^n u(-n-1)-\left(\frac{1-\sqrt{5}}{2}\right)^n u(n)\right]$$